Unit 1 – Using Objects and Methods Name: **KEY**
HW 8

1) What do the following commands evaluate to?

a) `Math.abs(5)` = 5

b) `Math.abs(-4.5)` = 4.5

c) `Math.pow(4, 3)` = 64.0

d) `Math.pow(0.5, 2)` = 0.25

e) `Math.pow(16, 0.5)` = 4.0

f) `Math.sqrt(16)` = 4.0

2) For the following expressions, give the range of values that the statement could produce.

`(int)(Math.random() * 10)`	a) 0 to 9
`(int)(Math.random() * 11)`	b) 0 to 10
`(int)(Math.random() * 100)`	c) 0 to 99
`(int)(Math.random() * 10) + 2`	d) 2 to 11
`(int)(Math.random() * 11) + 2`	e) 2 to 12
`(int)(Math.random() * 54) + 3`	f) 3 to 56

D 3) A pair of number cubes is used in a game of chance. Each number cube has 6 sides, numbered from 1 to 6 inclusive, and there is an equal probability for each of the numbers to appear on the top side (indicating the cube's value) when it is rolled. The following incomplete statement appears in a program that computes the sum of the values produced by rolling two number cubes.

```
int sum = /* missing code */
```

Which of the following replacements for `/* missing code */` would best simulate the value produced as a result of rolling two number cubes?

(A) `2 * (int) (Math.random() * 7)`
(B) `(int) (Math.random() * 6) + (int) (Math.random() * 6)`
(C) `(int) (Math.random() * 13)`
(D) `2 + (int) (Math.random() * 6) + (int) (Math.random() * 6)`

4) Rewrite the following formulas into valid Java expressions.

a) $\frac{|x-y|}{y^2}$

```
Math.abs(x - y)/Math.pow(y, 2)
```

b) $|a+b|^3 - 3a$

```
Math.pow(Math.abs(a+b), 3) - 3 * a
```

c) $\left(\frac{\sqrt{5+x}}{(y-2)^2}\right)^4$

```
Math.pow(Math.sqrt(5+x)/Math.pow(y - 2, 2), 4)
```

5) Write a single expression that takes the square root of 144 and 3 raised to the power of 3, multiplies them, then rounds the result to the nearest whole number.

```
(int) (Math.sqrt(144) * Math.pow(3, 3) + 0.5)
```

6) Write a single expression that generates a random integer between 0 and 9, raises that to the power of 2, and then divides the result by 4.

```
Math.pow((Math.random() * 10), 2) / 4
```

7) Write a single expression that computes the absolute difference between 2 to the power of 5 and –40.

```
Math.abs(Math.pow(2, 5) - (-40))
```

8) Write a single expression that raises 5 to the power of 3, takes the absolute value of –64, and then finds the remainder when dividing the larger of those two values by the smaller one.

```
Math.pow(5, 3) % Math.abs(-64)
```

9) Write a class called `QuadraticFormula`, which has three instance variables that represent the a, b, and c integer values in a quadratic equation: $ax^2 + bx + c = 0$. Create a default constructor that sets each of the values equal to 1. Have a parameter constructor. Create a method called `calculate1` that calculates the first x-intercept of a quadratic equation:

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Then, create a second method called `calculate2` that calculates the second x-intercept of a quadratic equation:

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Ignore the cases where you would take the square root of a negative; it should be assumed you would call on these methods to generate a real solution.

```
public class QuadraticFormula
{
  private int a;
  private int b;
  private int c;

  public QuadraticFormula() {
    a = 1;
    b = 1;
    c = 1;
  }

  public QuadraticFormula(int aa, int bb, int cc) {
    a = aa;
    b = bb;
    c = cc;
  }

  public double calculate1() {
    return ((-1 * b) + Math.sqrt(Math.pow(b, 2) - 4 * a *c))/(2*a);
  }

  public double calculate2() {
    return ((-1 * b) - Math.sqrt(Math.pow(b, 2) - 4 * a *c))/(2*a);
  }
}
```